

5(a)	force <u>per</u> unit positive charge	B1
5(b)(i)	$s = \frac{1}{2}at^2$	C1
	$a = (2 \times 0.045) / (1.5 \times 10^{-7})^2 = 4(.0) \times 10^{12} \text{ m s}^{-2}$	A1
5(b)(ii)	$F = 1.67 \times 10^{-27} \times 4.0 \times 10^{12} = 6.7 \text{ (6.68)} \times 10^{-15} \text{ N}$	A1
5(b)(iii)	1. $E = F / Q$	C1
	$= 6.68 \times 10^{-15} / 1.6 \times 10^{-19}$ $= 4.2 \text{ (4.18)} \times 10^4 \text{ NC}^{-1}$	A1
	2. $E = V / d$	C1
	$V = 4.18 \times 10^4 \times 0.045$ $= 1.9 \times 10^3 \text{ V}$	A1

Question	Answer	Marks
5(c)	$a = Eq / m$ or $F = ma$ and $F = Eq$	C1
	$\text{ratio} = \frac{(2 \times 1.6 \times 10^{-19}) \times (1.67 \times 10^{-27})}{(1.6 \times 10^{-19}) \times (4 \times 1.66 \times 10^{-27})} \text{ or } \frac{2 \times 1}{1 \times 4}$ $= 0.50$	A1

Question	Answer	Marks
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